

3-SAT Problem Overview

- A Boolean formula in conjunctive normal form (CNF) [Product of Sum]
 - Each clause has exactly 3 literals
- Goal: Is there a variable assignment that makes the formula true?

3-SAT Problem Overview

Example 1:

$$(A \vee B \vee C) \wedge (\sim A \vee \sim B \vee C) \wedge (A \vee \sim B \vee \sim C)$$

Solution:

A = True, B = False, C = True

Example2:

$$(A \vee A \vee A) \wedge (\sim A \vee \sim A \vee B) \wedge (\sim A \vee \sim A \vee \sim B)$$

Solution: ?

Clique Problem Overview

- A clique is a subset of vertices in a graph where every distinct pair of vertices within the subset is connected by an edge (A complete subgraph)
- Given a graph $G=(V,E)$, Find the maximum clique size. (Optimization variant)
- Given a graph $G=(V,E)$, does it contain a clique of size k ? (Decision variant)

From 3-SAT to Clique

Understanding the Reduction Process

Step 1: Define the Vertices

For Each Clause

Create a set of vertices.

Each Vertex = Literal

One vertex for each literal in the clause.

Total Vertices

3 vertices per clause, for m clauses: $3m$ total.

Example (for one clause)

Clause $(x_1 \vee \neg x_2 \vee x_3)$

3 vertices : x_1 , $\neg x_2$, and x_3

Step 2: Define the Edges

Different Clauses

Only connect vertices from different clauses.

Non-Complementary

Do not connect if literals are negations of each other.

Edge Example

x_1 and $\neg x_1$ not connected;
 x_1 and x_2 are connected
 x_2 and x_2 are also connected

Purpose

Ensures clique corresponds to consistent assignment.

Example

$$(x_2 + x_1 + \overline{x_3}) \cdot (\overline{x_1} + \overline{x_2} + x_4) \cdot (x_2 + \overline{x_4} + x_3)$$

$$\overline{x_1}$$

$$\overline{x_2}$$

$$x_4$$

$$x_2$$

$$x_2$$

$$x_1$$

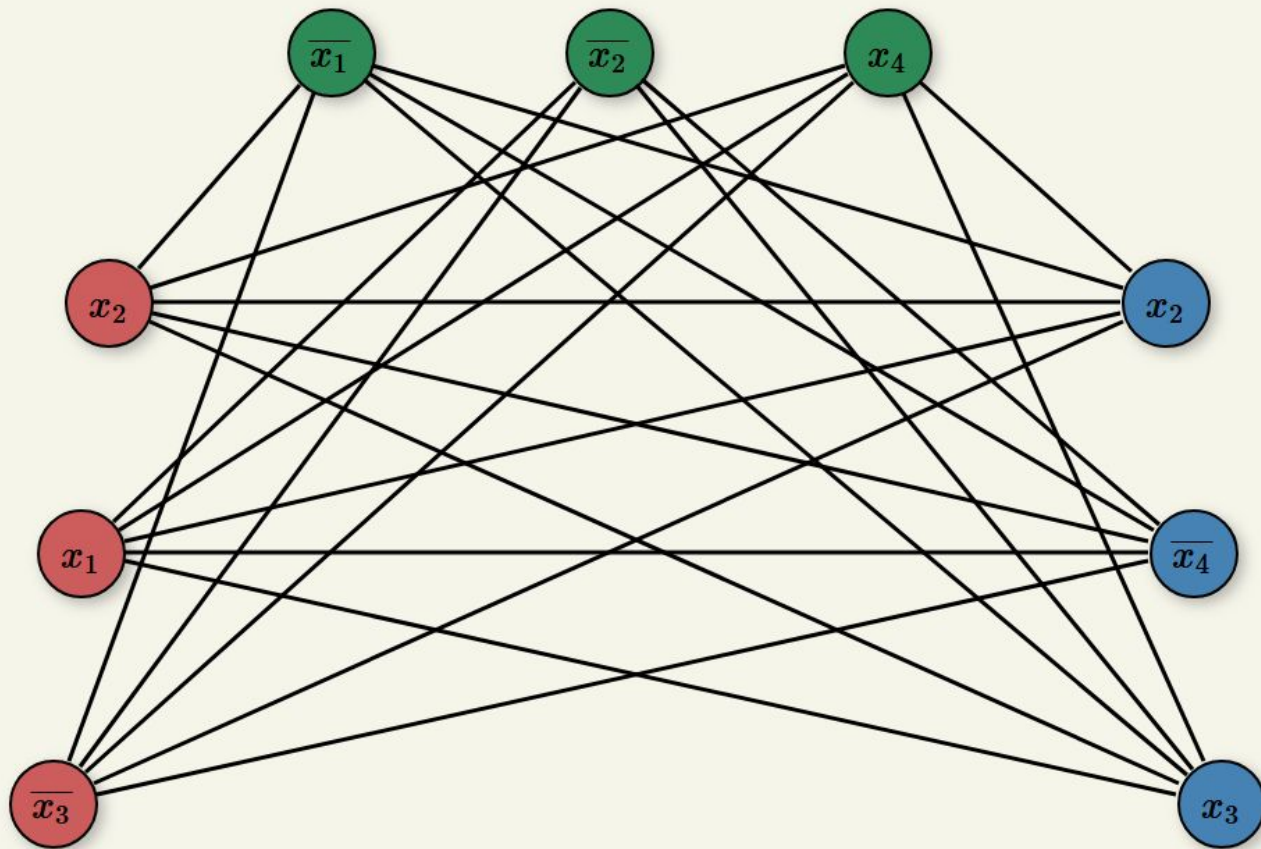
$$\overline{x_4}$$

$$\overline{x_3}$$

$$x_3$$

Example

$$(x_2 + x_1 + \overline{x_3}) \cdot (\overline{x_1} + \overline{x_2} + x_4) \cdot (x_2 + \overline{x_4} + x_3)$$



Why Does This Work?

Node Selection

A k -clique selects one literal from each clause.

No Contradictions

Edges only connect non-contradictory literals.

Satisfying Assignment

A clique implies a way to satisfy all clauses.

Equivalence

3-SAT is satisfied if all the vertices of the clique are assigned true value.

Vertex Cover Problem Overview

- Given a graph $G=(V,E)$
- Find a set C of vertices, so that every edge in E has at least one endpoint in C
- Optimization problem: Find the lowest possible value for $|C|$
- Decision problem: For a given value of k , is it possible to find $|C| \leq k$?



Building the Graph from 3-SAT

Variable Gadgets

For each variable x_i , create two vertices (v_i , not- v_i) connected by an edge.

Clause Gadgets

For each clause, create a triangle of three vertices, each for one literal.

Connecting Gadgets

Connect each clause vertex to its corresponding variable vertex.

Vertex Cover Size

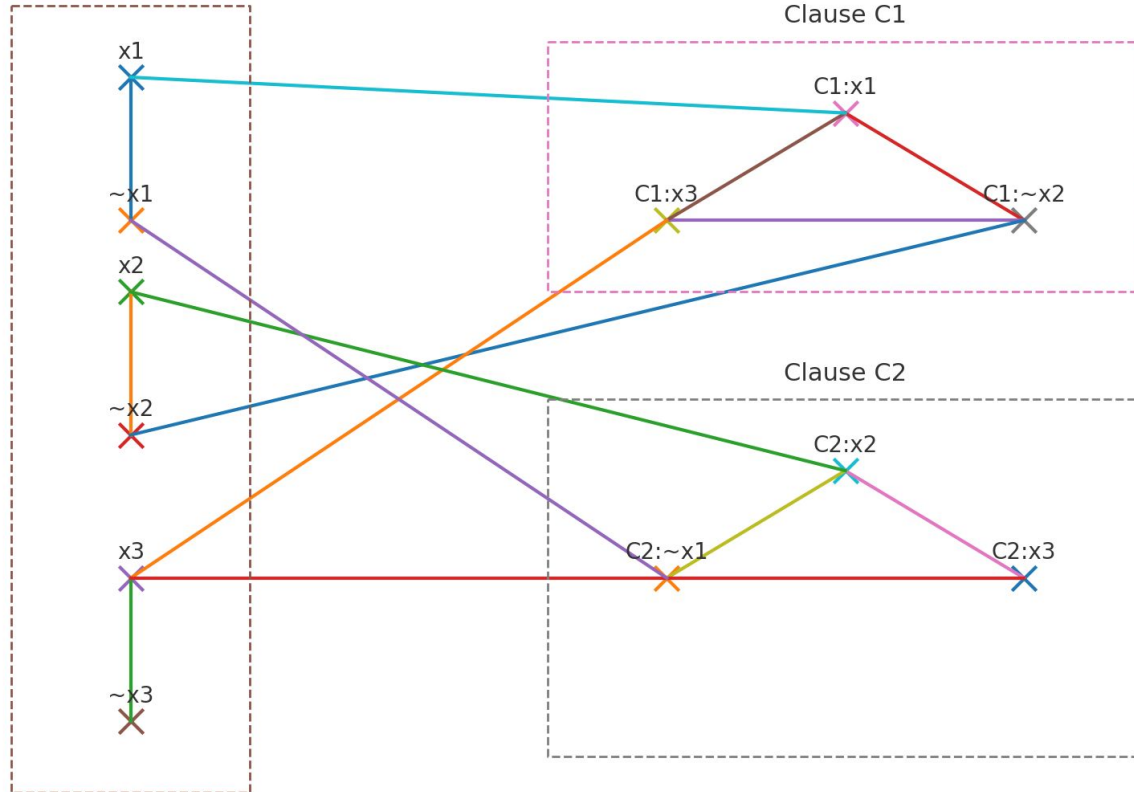
Set $k = n + 2m$, where n = variables, m = clauses.

Example

Reduction of 3-SAT to Vertex Cover

Example: $(x_1 \vee \neg x_2 \vee x_3) \wedge (x_2 \vee x_3 \vee \neg x_1)$ with $k = 7$

Variable gadgets



Why Does This Work?

If Satisfiable...

A satisfying assignment lets us pick one vertex per variable gadget and two per clause gadget, covering all edges with exactly k vertices.

If Vertex Cover Exists...

A cover of size k must pick one per variable pair and two per clause triangle, which corresponds to a satisfying assignment.



Converting Clique to 3-SAT

A Step-by-Step Reduction Guide

Step 1: Variable Construction

For each vertex v in V and each position i (1 to k), create a variable $x_{v,i}$. Meaning: 'vertex v is the i -th vertex in the clique.'

Step 2: Clause Construction

One Vertex for each position

For each position $i \in \{1, \dots, k\}$, at least one vertex should be chosen.

At most one vertex per position

For each position i , only one vertex can be chosen.

No Vertex Repeats

Each vertex appears at most once in the clique.

Clique Condition

There must be edge between chosen vertices

Clause Types and Examples

Clause Purpose	Clause Example
Atleast 3 vertex are selected	$\bigvee_{1 \leq i < j < k \leq n} (x_i \wedge x_j \wedge x_k)$
No 4 vertices can be choosen	$\bigwedge_{1 \leq i < j < k < l \leq n} (\neg x_i \vee \neg x_j \vee \neg x_k \vee \neg x_l)$
Selected vertices must be connected by edges. For every pair $(i, j) \notin E$	$ (\neg x_i \vee \neg x_j)$

Conversion to 3-CNF

Why Pad Clauses?

3-SAT requires each clause to have exactly 3 literals.

How to Pad?

- Any 2-CNF clause ($A \vee B$):
Can be written as: $(A \vee B \vee z) \wedge (A \vee B \vee \neg z)$
- Any clause having more than 3 literals ($L_1 \vee L_2 \vee \dots \vee L_k$)
Can be written as:
 $(L_1 \vee L_2 \vee z_1)$
 $(\neg z_1 \vee L_3 \vee z_2)$
 $(\neg z_2 \vee L_4 \vee z_3) \dots$
 $(\neg z_{k-3} \vee L_{k-1} \vee L_k)$